Toulouse School of Economics

Master 1 Thesis

Understanding 2.5-Degree Price Discrimination:

Imperfect Signals, Pricing, and Welfare

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Abstract

This paper studies *2.5-degree price discrimination*: a monopolist observes an imperfect public signal about consumer types and responds with a nonlinear menu, combining coarse segmentation with residual screening. Posterior virtual valuations satisfy a Bayes martingale property, so the signal is a mean-preserving spread of virtual values across signal realizations. Welfare can rise only when this spread triggers market shutdown for some type–signal pairs; otherwise total welfare falls. We decompose the welfare change into an expansion dividend and a variance penalty, and show that consumers bear twice the social variance cost while the firm earns a variance premium.

JEL Classification: D86, D82, D83, L11
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1 Introduction

Two classical models partition the economics of price discrimination along an information dimension. In second-degree discrimination, the seller knows only the population distribution of consumer types and elicits private information through a nonlinear tariff (Mussa and Rosen, 1978; Maskin and Riley, 1984): the instrument is rich but individual information is absent. In third-degree discrimination, the seller observes a perfect label—a verifiable demographic or geographic tag—and charges group-specific flat rates (Schmalensee, 1981; Varian, 1985): individual information exists but the instrument is blunt. Neither model captures a seller who has *both* an imperfect signal *and* a nonlinear menu—yet this combination increasingly characterizes real markets. Driven by the rapid proliferation of tracking technologies, firms now harvest granular individual-level data from a wide spectrum of sources—ranging from web cookies and telematics devices to credit bureaus (Acquisti et al., 2016)—and respond with differentiated contract schedules rather than posted prices. This paper studies that intermediate case, which we call *2.5-degree price discrimination* (imperfect public signal plus nonlinear screening within each signal realization), and examines its welfare implications.

We can see this motivating progression through the evolution of car insurance. Historically, when there was no signal to identify a driver's underlying risk, the insurer relied entirely on classical screening via a single nonlinear menu. Subsequently, insurers began gathering demographic information and verifiable driving records to better target their pricing strategies. Today, the landscape has shifted further: insurers now receive granular telematics data—acceleration patterns, braking frequency, and travel times—from third-party devices in the form of usage-based insurance scores. These data are informative but imperfect; they sort drivers into rough risk categories without perfectly revealing any individual's type. Within each category, a high-risk driver still knows more about their own riskiness than the insurer does. To prevent high-risk drivers from claiming the contracts designed for low-risk ones, the insurer must still leave informational rents to low-risk types. This is the standard second-degree distortion, now applied residually after the signal has done its partial sorting. The welfare question is therefore non-obvious: while more information can reduce the screening distortion (potentially re-opening markets for previously excluded consumers), an imperfect signal acts as a mean-preserving spread of virtual valuations across signal realizations. Crucially, such a spread disrupts the optimal allocation in ways the nonlinear menu cannot fully repair.

This institutional reality naturally raises a critical question regarding information design and data regulation. In practice, the precision of these signals is rarely determined by the principal (the insurer) alone; rather, it is engineered by third-party intermediaries—such as car manufacturers or data brokers—and is increasingly governed by social planners through privacy legislation. To understand how such entities should regulate or design this information structure, we must first unpack its fundamental economic consequences. Specifically, how does the introduction of an imperfect signal impact total social welfare, the firm's expected profit, and consumer surplus? Answering this question, and tracing how the surplus is reallocated among stakeholders, is the central focus of our analysis.

The welfare analysis of price discrimination has a long tradition. Schmalensee (1981) and Varian (1985) establish the foundational result for third-degree discrimination under linear pricing: welfare improves if and only if total output expands. Bergemann et al. (2015) characterize the complete welfare–surplus frontier achievable when a monopolist receives additional consumer information and responds with group-specific linear prices. We study the complementary case: the signal is imperfect and the firm responds with a nonlinear menu rather than a flat rate. This layering of coarse segmentation on top of within-group screening generates a welfare decomposition—expansion dividend versus variance penalty—that has no counterpart in the linear-pricing literature, and our market-shutdown condition is the natural nonlinear analogue of the output-expansion criterion.

Two related literatures clarify our contribution by contrast. Bergemann et al. (2022) characterize revenue-maximizing information disclosure in classic auctions: the seller chooses what bidders learn about their own valuations, and partial pooling of high types can be optimal. Eső and Szentes (2007) show that, in their auction setting where the seller releases additional signals to risk-neutral buyers, full disclosure of those signals can maximize expected revenue. We study a different environment on both margins: the signal is exogenous and observed by the *principal* about the agent's type, and the mechanism is nonlinear pricing rather than an auction. The welfare question therefore shifts from optimal disclosure design to the consequences of an exogenous mean-preserving spread of posterior virtual valuations. In this setting, total welfare can fall even when every type remains served under every signal realization—a channel absent when information flows from seller to bidders in auction formats analyzed for revenue maximization.

We study a principal-agent screening model in which a risk-neutral agent has a privately-known type $\theta \sim F$ on a bounded interval. An exogenous information structure generates a signal σ that is publicly observed by the principal, who offers a menu of contracts tailored to the realized posterior. The key analytical object is the *posterior virtual valuation*—the signal-adjusted version of the virtual type of Myerson (1981)—which governs the post-signal screening problem. Two discipline conditions are imposed: the Monotone Likelihood Ratio Property (MLRP), ensuring higher signals predict higher types, and a log-concavity condition on the posterior that rules out bunching.

Our first result is a martingale property: for every type θ, the prior virtual value equals the expectation of the posterior virtual value across signal realizations. The signal is therefore a mean-preserving spread of $v^\sigma(\theta)$ across σ for each fixed θ, and welfare consequences hinge on the curvature of the surplus function. A necessary condition for welfare gains follows: if every type is served under every signal realization, total welfare weakly falls (strictly so under an imperfect signal). Welfare can rise only when the signal is dispersed enough to shut down the market for some type–signal pairs, activating the convex region of the allocation rule. This shutdown condition is necessary but not sufficient: the welfare decomposition in Theorem 4.4 shows that the expansion dividend must dominate the variance penalty.

The distributional consequences are equally sharp. Decomposing the welfare change into an expansion dividend and a variance term reveals a striking asymmetry: the agent bears *twice* the variance penalty relative to society as a whole, while the firm's virtual profit—convex in the allocation because of the shutdown option—always generates a positive variance premium at each type. Before the agent learns θ or σ, expected virtual surplus coincides with expected actual surplus; integrating the pointwise decomposition therefore delivers the ex ante reallocation between consumers and the firm. The firm gains unambiguously from signal dispersion in expectation, while the agent may lose even when total welfare rises.

The remainder of the paper is organized as follows. Section 2 presents the model. Section 3 derives the martingale property of virtual valuations. Section 4 establishes the market-shutdown condition, the welfare decomposition, and—in Subsection 4.3—the distributional asymmetry between agent surplus and firm profit.

2 The Model

2.1 Primitives

We study the canonical nonlinear pricing model of Mussa and Rosen (1978) and Maskin and Riley (1984), enriched with an imperfect external signal. A risk-neutral principal sells a good to a risk-neutral agent whose type $\theta \in \Theta \equiv [\underline{\theta}, \bar{\theta}] \subset \mathbb{R}_+$ is private information. The prior distribution of θ has density f and cumulative distribution F, both continuously differentiable and strictly positive on Θ.

The agent's utility from consuming quantity $q \geq 0$ and paying transfer $T \in \mathbb{R}$ is

$$U(\theta, q, T) = \theta q - T,$$

so types are indexed by marginal willingness-to-pay and utility is linear in both quantity and the transfer, ruling out income effects. The principal's cost of producing quantity q is

$$c(q) = \tfrac{1}{2} k q^2, \qquad k > 0.$$

The quadratic specification delivers a linear first-best rule $q^*(\theta) = \theta/k$ and ensures that total surplus $S(\theta, q) \equiv \theta q - \frac{1}{2} k q^2$ is strictly concave in q for every θ—the curvature property on which the welfare analysis turns.

2.2 Information Structure

Before contracting, an exogenous signal $\sigma \in \Sigma$ about the agent's type is publicly observed by the principal. We assume that the signal is generated by the environment—telematics data, credit records, or similar external sources. Formally, it is described by a conditional density $\pi(\sigma \mid \theta)$, which together with the prior $f(\theta)$ induces a joint density

$$h(\theta, \sigma) = \pi(\sigma \mid \theta)\, f(\theta)$$

on $\Theta \times \Sigma$. The marginal density of σ is $g(\sigma) = \int_\Theta h(\theta, \sigma)\, d\theta$, and Bayes' rule delivers the posterior

$$f(\theta \mid \sigma) = \frac{h(\theta, \sigma)}{g(\sigma)}, \qquad F(\theta \mid \sigma) = \int_{\underline{\theta}}^{\theta} f(t \mid \sigma)\, dt.$$

Definition 2.1 (Imperfect signal)**.** The information structure is *imperfect*, or *non-degenerate*, if the signal does not perfectly reveal the agent's type: equivalently, the posterior $f(\theta \mid \sigma)$ remains non-degenerate for a set of signal realizations of positive measure, so residual screening within signal realizations is operative. Perfect revelation collapses the model to third-degree discrimination with type-specific first-best contracts and lies outside the maintained framework.

Two conditions are imposed on the information structure.

Assumption 2.2 (MLRP)**.** The joint density $h(\theta, \sigma)$ satisfies the strict monotone likelihood ratio property: for all $\theta_1 > \theta_2$ and $\sigma_1 > \sigma_2$,

$$\frac{h(\theta_1, \sigma_1)}{h(\theta_1, \sigma_2)} > \frac{h(\theta_2, \sigma_1)}{h(\theta_2, \sigma_2)}.$$

MLRP implies that a higher signal is good news about the agent's type in the likelihood ratio order. It preserves the monotonicity of the screening problem across signal realizations: conditional on any σ, higher types remain more profitable to serve.

Assumption 2.3 (Regularity)**.** For each $\sigma \in \Sigma$, the posterior $f(\theta \mid \sigma)$ is strictly log-concave in θ. Equivalently, the conditional inverse hazard rate

$$H(\theta \mid \sigma) \;\equiv\; \frac{1 - F(\theta \mid \sigma)}{f(\theta \mid \sigma)} \tag{1}$$

is strictly decreasing in θ.

Regularity is the standard single-crossing condition on the information rent. It ensures that the posterior virtual valuation $v^\sigma(\theta) \equiv \theta - H(\theta \mid \sigma)$ is strictly increasing in θ for every σ, so the principal's problem is globally well-behaved and the optimal allocation rule is free of bunching.

2.3 Timing

1. The information structure $\pi(\sigma \mid \theta)$ is fixed and commonly known.
2. Nature draws $\theta \sim F$; the signal $\sigma \sim \pi(\cdot \mid \theta)$ is realized and observed by the principal.
3. The principal offers a direct-revelation menu $\{(q(\hat{\theta}),\, T(\hat{\theta}))\}_{\hat{\theta}\in\Theta}$ to maximize expected profit given the posterior $f(\theta \mid \sigma)$.
4. The agent privately observes θ, selects a contract from the menu, and payoffs are realized.

The timing makes explicit the source of the residual screening problem. After σ is realized, the agent privately observes θ while the principal knows only the posterior $f(\theta \mid \sigma)$. Because the signal is imperfect, this posterior has non-degenerate support, and the principal must still leave informational rents to low types to prevent mimicry by higher types. It is precisely this residual rent—the standard second-degree distortion, now conditioned on σ—that drives the welfare analysis of the paper.

3 Screening and Virtual Valuations

3.1 The Optimal Contract

Given a signal realization σ, the principal faces a standard screening problem against the posterior $f(\theta \mid \sigma)$ derived in Section 2. By the revelation principle (Myerson, 1981), attention restricts without loss of generality to direct-revelation mechanisms $(q(\theta), T(\theta))_{\theta\in\Theta}$ satisfying incentive compatibility (IC) and individual rationality (IR). Standard integration by parts reduces the principal's problem to maximizing expected *virtual surplus* subject to a monotonicity constraint on q and the non-negativity constraint $q \geq 0$.

Define the conditional inverse hazard rate $H(\theta \mid \sigma)$ as in (1), and the corresponding *posterior virtual valuation*

$$v^{\sigma}(\theta) \;\equiv\; \theta - H(\theta \mid \sigma). \tag{2}$$

This is the signal-adjusted version of the virtual type of Myerson (1981): it discounts the agent's marginal value θ by the inverse hazard rate, which summarizes the information rent that must be left to lower types. Assumption 2.3 ensures $v^{\sigma}(\theta)$ is strictly increasing in θ for every σ, so the monotonicity constraint is non-binding and the optimal allocation solves the pointwise problem

$$\max_{q\geq 0}\, \left[v^{\sigma}(\theta) - kq\right] q,$$

yielding the closed-form rule

$$q^{\sigma}(\theta) = \max\left\{0,\, \frac{v^{\sigma}(\theta)}{k}\right\}. \tag{3}$$

The principal serves type θ under signal σ if and only if $v^{\sigma}(\theta) > 0$; otherwise the market shuts down for that type–signal pair. For later reference, define the *no-signal benchmark* as the allocation the principal would choose using only the prior $f(\theta)$:

$$v^{\phi}(\theta) \;\equiv\; \theta - H^{\phi}(\theta), \qquad H^{\phi}(\theta) \;\equiv\; \frac{1-F(\theta)}{f(\theta)}, \qquad q^{\phi}(\theta) \;=\; \max\left\{0,\, \frac{v^{\phi}(\theta)}{k}\right\}. \tag{4}$$

Throughout, the superscript ϕ denotes variables in the no-signal benchmark and σ denotes variables conditional on a realized signal.

3.2 The Martingale Property

The key structural result of this section is that, for each fixed type θ, the signal induces a *mean-preserving spread* of virtual valuations across signal realizations.

Lemma 3.1 (Martingale Property)**.** *For every $\theta \in \Theta$,*

$$\mathbb{E}[H(\theta \mid \sigma) \mid \theta] = H^{\phi}(\theta), \tag{5}$$

or equivalently, $\mathbb{E}[v^{\sigma}(\theta) \mid \theta] = v^{\phi}(\theta)$.

Proof. Fix θ. By Bayes' rule, $f(\sigma \mid \theta) = h(\theta, \sigma)/f(\theta)$. Hence

$$\mathbb{E}[H(\theta \mid \sigma) \mid \theta] = \frac{1}{f(\theta)} \int_{\Sigma} H(\theta \mid \sigma)\, f(\theta \mid \sigma)\, g(\sigma)\, d\sigma.$$

Since $H(\theta \mid \sigma)\, f(\theta \mid \sigma) = 1 - F(\theta \mid \sigma)$ by definition of H, we have

$$\mathbb{E}[H(\theta \mid \sigma) \mid \theta] = \frac{1}{f(\theta)} \int_{\Sigma} [1 - F(\theta \mid \sigma)]\, g(\sigma)\, d\sigma.$$

Let $\tilde{\theta}$ denote a random type drawn from the prior. Iterated expectations give

$$\int_{\Sigma} F(\theta \mid \sigma)\, g(\sigma)\, d\sigma = \mathbb{E}_{\sigma}[\mathbb{P}(\tilde{\theta} \le \theta \mid \sigma)] = \mathbb{P}(\tilde{\theta} \le \theta) = F(\theta),$$

so $\mathbb{E}[H(\theta \mid \sigma) \mid \theta] = [1 - F(\theta)]/f(\theta) = H^{\phi}(\theta)$. The statement for virtual values follows from linearity: $v^{\sigma}(\theta) = \theta - H(\theta \mid \sigma)$. □

Lemma 3.1 has a transparent economic interpretation. The information structure $\pi(\sigma \mid \theta)$ redistributes probability mass across posteriors but, by Bayes' law, leaves the marginal distribution of types unchanged—a property widely formalized in the information design literature as Bayesian plausibility (Kamenica and Gentzkow, 2011). Because the information rent $H(\theta \mid \sigma)$ is a function of the posterior alone, its cross-signal expectation must equal its prior value. For each fixed θ, the signal therefore raises the dispersion of $v^{\sigma}(\theta)$ around its mean $v^{\phi}(\theta)$ without shifting that mean.

This observation is the analytical foundation for all welfare results that follow. Since the optimal allocation (3) is a convex function of the virtual value, Jensen's inequality immediately implies that signal dispersion weakly *expands* expected output for every type. Whether this output expansion improves welfare depends on where in the type distribution the expansion occurs— the question taken up in Section 4.

4 Welfare Analysis

This section characterizes the welfare consequences of the signal. We compare the signal regime—in which the principal observes σ and screens against the posterior $f(\theta \mid \sigma)$—against the no-signal benchmark defined in (4). Define the pointwise welfare change

$$\Delta W(\theta) \;\equiv\; \mathbb{E}_{\sigma|\theta}[S(\theta,\, q^{\sigma}(\theta))] - S\Big(\theta,\, q^{\phi}(\theta)\Big)\,, \tag{6}$$

so that the aggregate welfare change is $\Delta TW = \int_{\Theta} \Delta W(\theta)\, f(\theta)\, d\theta$.

4.1 The Market-Shutdown Condition

The martingale property (Lemma 3.1) establishes that, for each θ, the signal is a mean-preserving spread of $v^\sigma(\theta)$ across σ. The welfare consequences of this spread depend critically on the shape of the indirect surplus function. When every consumer is served under every signal realization, the allocation is everywhere interior and the indirect surplus $W^{\text{ind}}(v;\theta)$ is strictly concave in v—a mean-preserving spread then reduces welfare by Jensen's inequality. The non-negativity constraint on quantity is what opens the door to welfare gains: if the signal is dispersed enough to drive some posterior virtual values below zero, the principal shuts down the market for those type–signal pairs, and the truncation at zero introduces a locally convex region that can overturn the Jensen penalty.

Lemma 4.1 (No welfare gain without market shutdown)**.** *Suppose Assumptions 2.2 and 2.3 hold and the information structure is imperfect in the sense of Definition 2.1. If $q^\sigma(\underline{\theta}) > 0$ for all $\sigma \in \Sigma$, then*

$$\Delta TW \;\leq\; 0,$$

with strict inequality whenever $\mathrm{Var}[v^\sigma(\theta) \mid \theta] > 0$ *on a set of positive measure.*

Proof. The premise $q^\sigma(\underline{\theta}) > 0$ requires $v^\sigma(\underline{\theta}) > 0$ for all $\sigma \in \Sigma$. By Assumption 2.3, $v^\sigma(\theta)$ is strictly increasing in θ, so $v^\sigma(\theta) > 0$ for all $(\theta, \sigma) \in \Theta \times \Sigma$. Global positivity implies the non-negativity constraint is everywhere slack, and the optimal allocation is always the interior solution $q^\sigma(\theta) = v^\sigma(\theta)/k$.

Substituting into $S(\theta, q) = \theta q - \frac{1}{2}kq^2$, the indirect welfare function is

$$W^{\text{ind}}(v^\sigma;\, \theta) \;=\; S\Big(\theta,\, \frac{v^\sigma}{k}\Big) \;=\; \frac{1}{k}\Big(\theta v^\sigma - \frac{1}{2}(v^\sigma)^2\Big).$$

This is strictly concave in v^σ, since $\partial^2 W^{\text{ind}}/\partial(v^\sigma)^2 = -1/k < 0$.

The martingale property (Lemma 3.1) gives $\mathbb{E}_{\sigma|\theta}[v^\sigma(\theta)] = v^\phi(\theta)$ for every θ. Applying Jensen's inequality to the strictly concave function $W^{\text{ind}}(\,\cdot\,;\theta)$:

$$\mathbb{E}_{\sigma|\theta}\Big[W^{\text{ind}}(v^\sigma(\theta);\, \theta)\Big] \;\leq\; W^{\text{ind}}\Big(\mathbb{E}_{\sigma|\theta}[v^\sigma(\theta)];\, \theta\Big) \;=\; W^{\text{ind}}(v^\phi(\theta);\, \theta),$$

with strict inequality when $\mathrm{Var}[v^\sigma(\theta) \mid \theta] > 0$. Since $\Delta W(\theta) = \mathbb{E}_{\sigma|\theta}[W^{\text{ind}}(v^\sigma;\theta)] - W^{\text{ind}}(v^\phi;\theta) \leq 0$ for every θ, integrating over Θ yields $\Delta TW \leq 0$. □

Remark 4.2. Lemma 4.1 applies to imperfect signals satisfying Assumption 2.3. Perfect revelation lies outside this framework: the posterior collapses to a point mass, screening rents vanish ($H(\theta \mid \sigma) \to 0$), and the first-best allocation $q^*(\theta) = \theta/k$ is restored. The lemma therefore establishes that, within the class of imperfect signals, welfare cannot rise unless the signal triggers market shutdown for some type–signal pairs.

The converse direction is equally important. The condition $q^\phi(\underline{\theta}) > 0$—the market being active under the prior—does *not* suffice to guarantee $\Delta TW \leq 0$. Even if no type is excluded in the benchmark, a sufficiently dispersed signal can induce negative posterior virtual values for some realizations, activating the shutdown region and potentially generating a net welfare gain. The relevant condition is whether shutdown occurs *under the signal*, not under the prior. Shutdown is necessary for welfare gains but not sufficient: Theorem 4.4 shows that the expansion dividend must also dominate the variance penalty.

4.2 The Welfare Decomposition

To characterize the sign and magnitude of ΔTW in the general case, we first establish that the convexity of the optimal allocation rule always produces a non-negative quantity expansion in expectation.

Lemma 4.3 (Quantity expansion)**.** *For any $\theta \in \Theta$, define the expected quantity expansion*

$$\Delta\bar{q}(\theta) \;\equiv\; \mathbb{E}_{\sigma|\theta}[q^\sigma(\theta)] - q^\phi(\theta).$$

Then $\Delta\bar{q}(\theta) \geq 0$ for all $\theta \in \Theta$, with strict inequality if and only if the support of $v^\sigma(\theta)$ straddles zero.

Proof. The optimal allocation rule (3) can be written as $q(v) = \max\{0, v/k\}$, which is a convex function of v. The martingale property gives $\mathbb{E}_{\sigma|\theta}[v^\sigma(\theta)] = v^\phi(\theta)$. Applying Jensen's inequality to the convex function $q(\cdot)$:

$$\mathbb{E}_{\sigma|\theta}[q^\sigma(\theta)] = \mathbb{E}_{\sigma|\theta}[q(v^\sigma(\theta))] \geq q\Big(\mathbb{E}_{\sigma|\theta}[v^\sigma(\theta)]\Big) = q(v^\phi(\theta)) = q^\phi(\theta).$$

The inequality is strict if and only if $q(\cdot)$ is not affine on the support of $v^\sigma(\theta)$, which occurs precisely when the support straddles the kink at $v = 0$. □

Theorem 4.4 (Welfare decomposition)**.** *Under Assumptions 2.2 and 2.3, for every $\theta \in \Theta$,*

$$\Delta W(\theta) \;=\; \underbrace{\left[\theta - \frac{k}{2}\Big(\mathbb{E}_{\sigma|\theta}[q^\sigma(\theta)] + q^\phi(\theta)\Big)\right]}_{\textit{average net marginal surplus}} \cdot \underbrace{\Delta\bar{q}(\theta)}_{\textit{expansion dividend}} - \underbrace{\frac{k}{2}\,\mathrm{Var}[q^\sigma(\theta) \mid \theta]}_{\textit{variance penalty}}. \tag{7}$$

Consequently, $\Delta TW > 0$ if and only if

$$\int_\Theta \left[\theta - \frac{k}{2}\Big(\mathbb{E}_{\sigma|\theta}[q^\sigma] + q^\phi\Big)\right] \Delta\bar{q}(\theta)\, f(\theta)\, d\theta \;>\; \frac{k}{2}\int_\Theta \mathrm{Var}[q^\sigma \mid \theta]\, f(\theta)\, d\theta. \tag{8}$$

Proof. By definition (6) and the surplus function $S(\theta, q) = \theta q - \frac{k}{2}q^2$,

$$\Delta W(\theta) = \theta\left(\mathbb{E}_{\sigma|\theta}[q^\sigma] - q^\phi\right) - \frac{k}{2}\left(\mathbb{E}_{\sigma|\theta}[(q^\sigma)^2] - (q^\phi)^2\right).$$

We decompose the quadratic term using the variance identity $\mathbb{E}[(q^\sigma)^2] = \mathrm{Var}[q^\sigma \mid \theta] + (\mathbb{E}[q^\sigma])^2$:

$$\begin{aligned}
\mathbb{E}_{\sigma|\theta}[(q^\sigma)^2] - (q^\phi)^2 &= \mathrm{Var}[q^\sigma \mid \theta] + \left(\mathbb{E}_{\sigma|\theta}[q^\sigma]\right)^2 - (q^\phi)^2 \\
&= \mathrm{Var}[q^\sigma \mid \theta] + \underbrace{\left(\mathbb{E}_{\sigma|\theta}[q^\sigma] - q^\phi\right)}_{\Delta\bar{q}(\theta)}\left(\mathbb{E}_{\sigma|\theta}[q^\sigma] + q^\phi\right).
\end{aligned}$$

Substituting back and factoring $\Delta\bar{q}(\theta)$:

$$\begin{aligned}
\Delta W(\theta) &= \theta\,\Delta\bar{q}(\theta) - \frac{k}{2}\Big[\mathrm{Var}[q^\sigma \mid \theta] + \Delta\bar{q}(\theta)\left(\mathbb{E}_{\sigma|\theta}[q^\sigma] + q^\phi\right)\Big] \\
&= \left[\theta - \frac{k}{2}\Big(\mathbb{E}_{\sigma|\theta}[q^\sigma] + q^\phi\Big)\right]\Delta\bar{q}(\theta) - \frac{k}{2}\,\mathrm{Var}[q^\sigma \mid \theta].
\end{aligned}$$

Integrating over Θ with weight $f(\theta)$ yields (8). □

The decomposition (7) has a transparent geometric interpretation, illustrated in Figure 1 for a fixed type θ. The signal raises the *expected* allocation from q^ϕ to $\mathbb{E}_{\sigma|\theta}[q^\sigma]$; the social gain from this expansion is the area of the green trapezoid bounded above by the marginal benefit θ and below by the average marginal cost $\frac{k}{2}(\mathbb{E}[q^\sigma] + q^\phi)$—the red arrow in the figure. The variance penalty $\frac{k}{2}\mathrm{Var}[q^\sigma \mid \theta]$ has no counterpart in this diagram: it reflects allocation volatility *within* the active region across signal realizations, not a shift of the boundary. The connection to Lemma 4.1 is then immediate: when no type is ever excluded, Lemma 4.3 implies $\Delta\bar{q}(\theta) = 0$ for every θ, the trapezoid collapses to zero area, and only the variance penalty survives.

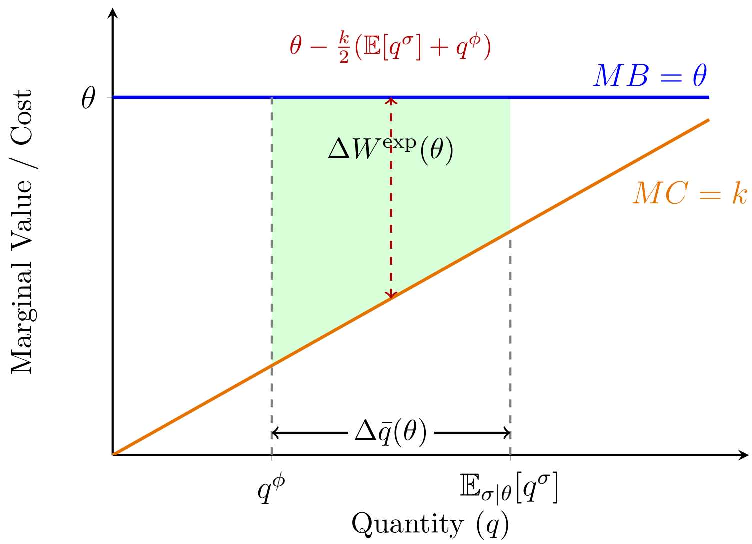

Figure 1: Geometric representation of the expansion dividend in (7) for a fixed type θ. The signal raises expected output from q^ϕ to $\mathbb{E}_{\sigma|\theta}[q^\sigma]$. The social value of this expansion equals the area of the green trapezoid: the quantity expansion $\Delta\bar{q}(\theta)$ multiplied by the average net marginal surplus $\theta - \frac{k}{2}(\mathbb{E}[q^\sigma] + q^\phi)$ (red arrow). The variance penalty $\frac{k}{2}\mathrm{Var}[q^\sigma \mid \theta]$ is not depicted, as it operates within the active region and carries no geometric footprint in this diagram.

Figure 2 provides a numerical illustration of the welfare decomposition (7). At each type θ, green shading marks the expansion dividend and red shading the variance penalty; thin blue and red curves overlay virtual agent surplus and principal profit. The accompanying code is described in Appendix.

4.3 Distributional Consequences

The aggregate welfare change masks a sharp asymmetry in how the signal's effects are distributed between the two parties. For a given signal realization σ, let $U^\sigma(\theta) \equiv \theta q^\sigma(\theta) - T^\sigma(\theta)$ denote the agent's *actual* information rent and $\Pi^\sigma(\theta) \equiv T^\sigma(\theta) - \frac{k}{2}(q^\sigma(\theta))^2$ denote the principal's *actual* profit. Define the *virtual agent surplus* and *virtual profit* as the accounting objects

$$\widetilde{AS}(\theta \mid \sigma) \;\equiv\; H(\theta \mid \sigma)\, q^\sigma(\theta), \qquad \widetilde{\Pi}(\theta \mid \sigma) \;\equiv\; S(\theta, q^\sigma(\theta)) - \widetilde{AS}(\theta \mid \sigma) = \frac{k}{2}\big(q^\sigma(\theta)\big)^2.$$

Under an interior allocation, $H(\theta \mid \sigma) = \theta - kq^\sigma(\theta)$, so $\widetilde{AS}(\theta \mid \sigma) = \theta q^\sigma(\theta) - k(q^\sigma(\theta))^2$. At shutdown ($q^\sigma = 0$), both actual and virtual surplus equal zero. The virtual objects are not equal to actual payoffs *type by type*, but they coincide in expectation behind a veil of ignorance.

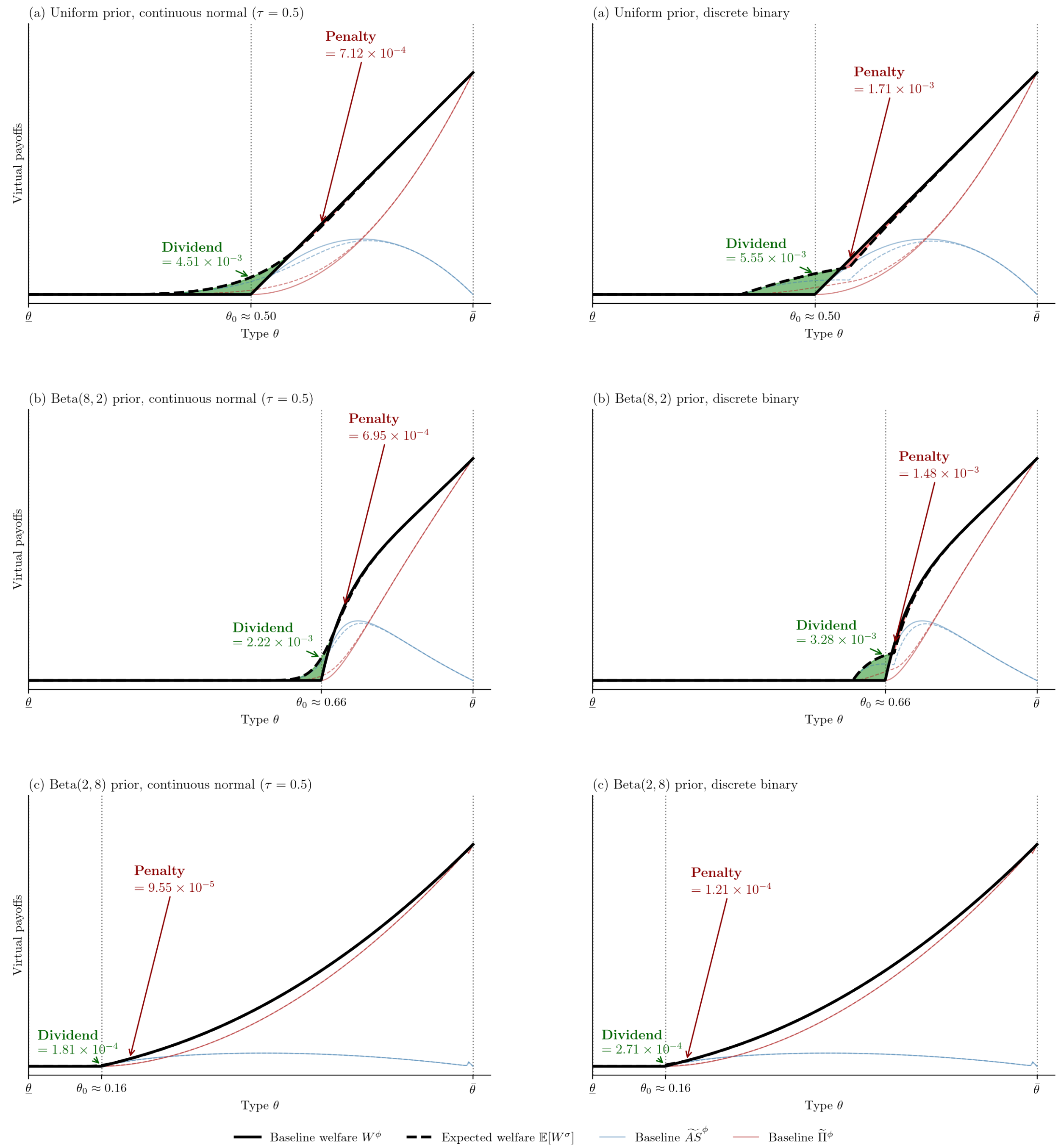

Figure 2: Numerical illustration of the welfare decomposition (7) in a 3×2 panel layout. Rows use, from top to bottom, a uniform prior, a right-skewed Beta$(8, 2)$ prior, and a left-skewed Beta$(2, 8)$ prior. Columns use continuous Gaussian noise ($\tau = 0.5$) and a discrete binary signal. Bold black curves plot baseline and expected total welfare; thin blue and red curves plot virtual agent surplus and principal profit. Green (red) shading marks the expansion dividend (variance penalty). See Appendix.

For the no-signal benchmark, write $\widetilde{AS}^{\phi}(\theta) \equiv H^{\phi}(\theta)\, q^{\phi}(\theta)$ and $\widetilde{\Pi}^{\phi}(\theta) \equiv \frac{k}{2}(q^{\phi}(\theta))^2$. The pointwise changes in virtual agent surplus and virtual profit are

$$\Delta\widetilde{AS}(\theta) \;\equiv\; \mathbb{E}_{\sigma|\theta}\Big[\widetilde{AS}(\theta \mid \sigma)\Big] - \widetilde{AS}^{\phi}(\theta), \qquad \Delta\widetilde{\Pi}(\theta) \;\equiv\; \mathbb{E}_{\sigma|\theta}\Big[\widetilde{\Pi}(\theta \mid \sigma)\Big] - \widetilde{\Pi}^{\phi}(\theta). \tag{9}$$

Lemma 4.5 (Ex ante surplus identities)**.** *Suppose Assumptions 2.2 and 2.3 hold and IR binds at $\underline{\theta}$ in each regime. Then, for the no-signal benchmark,*

$$\int_{\Theta} U^{\phi}(\theta)\, f(\theta)\, d\theta = \int_{\Theta} \widetilde{AS}^{\phi}(\theta)\, f(\theta)\, d\theta, \qquad \int_{\Theta} \Pi^{\phi}(\theta)\, f(\theta)\, d\theta = \int_{\Theta} \widetilde{\Pi}^{\phi}(\theta)\, f(\theta)\, d\theta,$$

Under the signal regime,

$$\int_{\Theta}\int_{\Sigma} U^{\sigma}(\theta)\, h(\theta,\sigma)\, d\sigma\, d\theta = \int_{\Theta}\int_{\Sigma} \widetilde{AS}(\theta \mid \sigma)\, h(\theta,\sigma)\, d\sigma\, d\theta,$$
$$\int_{\Theta}\int_{\Sigma} \Pi^{\sigma}(\theta)\, h(\theta,\sigma)\, d\sigma\, d\theta = \int_{\Theta}\int_{\Sigma} \widetilde{\Pi}(\theta \mid \sigma)\, h(\theta,\sigma)\, d\sigma\, d\theta.$$

Proof. Fix σ. Under the optimal IC contract with $U^{\sigma}(\underline{\theta}) = 0$, the envelope theorem gives $U^{\sigma}(\theta) = \int_{\underline{\theta}}^{\theta} q^{\sigma}(t)\, dt$. Integration by parts against the posterior $f(\theta \mid \sigma)$ yields

$$\int_{\Theta} U^{\sigma}(\theta)\, f(\theta \mid \sigma)\, d\theta = \int_{\Theta} H(\theta \mid \sigma)\, q^{\sigma}(\theta)\, f(\theta \mid \sigma)\, d\theta = \int_{\Theta} \widetilde{AS}(\theta \mid \sigma)\, f(\theta \mid \sigma)\, d\theta.$$

Multiplying by $g(\sigma)$ and integrating over Σ gives the signal-regime identity for the agent; the benchmark case is the special case with a degenerate signal. For profit, $\Pi^{\sigma}(\theta) = S(\theta, q^{\sigma}(\theta)) - U^{\sigma}(\theta)$, so integrating and applying the agent identity gives $\int \Pi^{\sigma} f(\cdot \mid \sigma)\, d\theta = \int \widetilde{\Pi}(\theta \mid \sigma) f(\cdot \mid \sigma)\, d\theta$ for each σ, and integrating over σ yields the second pair of identities. $\square$

Remark 4.6. Lemma 4.5 justifies interpreting the virtual objects from the ex ante perspective, before the agent learns θ or σ. The pointwise changes in (9) are not type-by-type changes in actual rent, but their prior-weighted integrals coincide with the ex ante changes in expected actual surplus and profit:

$$\int_{\Theta} \Delta\widetilde{AS}(\theta)\, f(\theta)\, d\theta = \Delta\mathbb{E}[U], \qquad \int_{\Theta} \Delta\widetilde{\Pi}(\theta)\, f(\theta)\, d\theta = \Delta\mathbb{E}[\Pi],$$

where expectations are taken behind the veil of ignorance over θ and σ.

Corollary 4.7 (Distributional decomposition)**.** *Under Assumptions 2.2 and 2.3, the pointwise changes in* (9) *satisfy*

$$\Delta\widetilde{AS}(\theta) \;=\; \Big[\theta - k\Big(\mathbb{E}_{\sigma|\theta}[q^{\sigma}] + q^{\phi}\Big)\Big]\, \Delta\bar{q}(\theta) \;-\; k\, \mathrm{Var}[q^{\sigma}(\theta) \mid \theta], \tag{10}$$

$$\Delta\widetilde{\Pi}(\theta) \;=\; \frac{k}{2}\Big(\mathbb{E}_{\sigma|\theta}[q^{\sigma}] + q^{\phi}\Big)\Delta\bar{q}(\theta) \;+\; \frac{k}{2}\, \mathrm{Var}[q^{\sigma}(\theta) \mid \theta]. \tag{11}$$

Consequently, $\Delta\widetilde{\Pi}(\theta) \geq 0$ for every θ, so ex ante expected profit weakly increases: $\int_{\Theta} \Delta\widetilde{\Pi}(\theta)\, f(\theta)\, d\theta = \Delta\mathbb{E}[\Pi] \geq 0$.

Proof. For (10), compute

$$\Delta\widetilde{AS}(\theta) = \mathbb{E}_{\sigma|\theta}\Big[\theta q^{\sigma} - k(q^{\sigma})^2\Big] - \Big(\theta q^{\phi} - k(q^{\phi})^2\Big).$$

Separating the linear and quadratic terms:

$$\Delta\widetilde{AS}(\theta) = \theta\Delta\bar{q}(\theta) - k\left(\mathbb{E}_{\sigma|\theta}[(q^\sigma)^2] - (q^\phi)^2\right).$$

Applying the same variance decomposition as in the proof of Theorem 4.4,

$$\mathbb{E}_{\sigma|\theta}[(q^\sigma)^2] - (q^\phi)^2 = \mathrm{Var}[q^\sigma \mid \theta] + \Delta\bar{q}(\theta)\left(\mathbb{E}_{\sigma|\theta}[q^\sigma] + q^\phi\right),$$

and factoring $\Delta\bar{q}(\theta)$ yields (10).

For (11), compute

$$\Delta\widetilde{\Pi}(\theta) = \frac{k}{2}\left(\mathbb{E}_{\sigma|\theta}[(q^\sigma)^2] - (q^\phi)^2\right) = \frac{k}{2}\left[\mathrm{Var}[q^\sigma \mid \theta] + \Delta\bar{q}(\theta)\left(\mathbb{E}_{\sigma|\theta}[q^\sigma] + q^\phi\right)\right],$$

which rearranges to (11). Internal consistency is verified by noting that (10) + (11) = (7). Both terms on the right-hand side of (11) are non-negative, so $\Delta\widetilde{\Pi}(\theta) \geq 0$. □

Comparing (10) with (7) reveals the asymmetry precisely. The variance coefficient in the agent's decomposition is k—exactly *twice* the social variance penalty $\frac{k}{2}$. The agent therefore absorbs twice the second-order cost of allocation volatility relative to the social optimum. The firm, by contrast, earns a strict *variance premium* $+\frac{k}{2}\mathrm{Var}[q^\sigma \mid \theta] \geq 0$ in (11): the virtual profit function $\widetilde{\Pi} \propto q^2$ is convex in the allocation, so Jensen's inequality works in the firm's favor. By Remark 4.6, integrating these pointwise expressions yields the ex ante conclusion: the firm gains unambiguously from signal dispersion in expectation, while the agent may lose even when total welfare rises. Data tracking can therefore be simultaneously welfare-improving for society and welfare-reducing for the average consumer.

5 Conclusion

This paper has studied the welfare consequences of imperfect external signals in a nonlinear pricing environment—a regime we term 2.5-degree price discrimination, lying between the classical second- and third-degree benchmarks. The central analytical object is the posterior virtual valuation, which inherits a martingale property from Bayes' law: for every type θ, the prior virtual value equals the expectation of its posterior counterpart across all signal realizations. For each fixed θ, the signal is therefore a mean-preserving spread of virtual valuations across σ, and all welfare results follow from the interplay between this spread and the curvature of the surplus function.

Three findings emerge from the analysis. First, the market-shutdown condition (Lemma 4.1) identifies a *necessary* prerequisite for welfare gains: if every type is served under every signal realization, total welfare weakly falls. Welfare can rise only when shutdown occurs for some type–signal pairs, though shutdown alone is not sufficient—the expansion dividend must dominate the variance penalty (Theorem 4.4). Second, the welfare decomposition separates the aggregate welfare change into an expansion dividend and a variance penalty from allocation volatility. Whether the signal improves or reduces welfare hinges on which term dominates. Third, the distributional decomposition (Corollary 4.7), together with the ex ante identities in Lemma 4.5, reveals a structural asymmetry: the agent bears twice the social variance penalty in expectation, while the firm earns an unambiguous variance premium. Data tracking therefore reallocates surplus from consumers to the firm even in markets where it raises total welfare.

Directions for Future Research

The model is deliberately parsimonious, and several extensions are both natural and economically important. We briefly outline four directions below.

A first extension is to endogenize the signal when private parties control its design. Throughout this paper, $\pi(\sigma \mid \theta)$ is exogenous—generated by the environment rather than by either party. A natural next step is to ask which party would design the signal if given the choice, and what structure each would prefer. The distributional decomposition suggests that the principal gains unambiguously from richer signals that induce mean-preserving spreads of posterior virtual valuations and would therefore favor maximal dispersion subject to regularity, in the spirit of Bayesian persuasion (Kamenica and Gentzkow, 2011). The agent faces the opposite trade-off: a larger variance penalty, offset only when the expansion dividend is positive. Characterizing these private design problems is conceptually clear but technically demanding. The feasible set of information structures is infinite-dimensional—every prior on Θ combined with every conditional law $\pi(\sigma \mid \theta)$ satisfying MLRP and regularity defines a distinct screening problem—and optimal design over this space calls for the functional-analytic tools of modern information design.

A closely related, but distinct, policy question arises when a social planner observes an imperfect signal and chooses what to disclose to the principal in order to maximize total welfare. This is not the same as the firm's profit-maximizing design problem: the planner internalizes the expansion dividend and the variance penalty with social weights, and may optimally withhold or garble information that a profit-maximizing principal would prefer to amplify. The martingale property of Lemma 3.1 still constrains any feasible disclosure rule—virtual valuations must remain a mean-preserving spread—but the planner's objective selects a different point on the same feasible frontier. Because both the prior and the signal space can vary without bound across applications, a general closed-form solution is unlikely; progress would require either additional structure on admissible signals (e.g., deterministic partitions or upper censorship), linear-programming approaches to optimal disclosure (Kolotilin, 2018), or the numerical methods described in Appendix.

The present model also assumes that the signal is publicly observed by the principal alone after the agent's type is realized. A richer setting would allow the agent to hold a private signal about their own type before contracting, generating bilateral asymmetric information. In such a setting, the agent's information rent depends on both the principal's posterior and the agent's private belief, and the interaction between the two information structures can generate qualitatively new effects on screening and welfare.

Finally, the welfare results derived here apply to a monopolist. Extending the analysis to competing firms would introduce a second margin: imperfect signals not only affect the screening distortion within each firm, but also shape how consumers sort across firms. Whether competition amplifies or dampens the welfare effects of data tracking is an open question with direct policy relevance, particularly in platform markets where firms share access to consumer data through common intermediaries such as data brokers.

Closing Remarks

This thesis suggests that the intermediate regime of *2.5-degree price discrimination*—imperfect signals combined with within-group nonlinear screening—opens a research agenda centered on a practical question: *who* designs or processes the signal, and *how*? In practice, the signal may be engineered by a data broker, shaped by a social planner through privacy or disclosure rules, or chosen by the principal to maximize profit; each party faces a different objective, yet all operate within the same structural constraint that signals redistribute virtual valuations without shifting their mean. The framework developed here makes the downstream consequences of any such design choice transparent—how total welfare, consumer surplus, and firm profit respond through the expansion dividend and the variance penalty. These are precisely the quantities a social planner weighs when regulating data markets, a data broker evaluates when selling scoring products, and a principal and agent each internalize when negotiating contracts under residual private information. Fully characterizing optimal signal design or processing along these lines is far from straightforward: the feasible set of information structures is rich, objectives conflict across stakeholders, and institutional details matter. But by isolating the welfare logic of imperfect signals in a nonlinear pricing environment, this paper offers a foundation on which that harder, and economically important, program of research can build.

References

Acquisti, A., Taylor, C., and Wagman, L. (2016). The economics of privacy. *Journal of economic Literature*, 54(2):442–492.

Bergemann, D., Brooks, B., and Morris, S. (2015). The limits of price discrimination. *American Economic Review*, 105(3):921–957.

Bergemann, D., Heumann, T., Morris, S., Sorokin, C., and Winter, E. (2022). Optimal information disclosure in classic auctions. *American Economic Review: Insights*, 4(3):371–388.

Eső, P. and Szentes, B. (2007). Optimal information disclosure in auctions and the handicap auction. *The Review of Economic Studies*, 74(3):705–731.

Kamenica, E. and Gentzkow, M. (2011). Bayesian persuasion. *American Economic Review*, 101(6):2590–2615.

Kolotilin, A. (2018). Optimal information disclosure: A linear programming approach. *Theoretical Economics*, 13(2):607–636.

Maskin, E. and Riley, J. (1984). Monopoly with incomplete information. *The RAND Journal of Economics*, 15(2):171–196.

Mussa, M. and Rosen, S. (1978). Monopoly and product quality. *Journal of Economic theory*, 18(2):301–317.

Myerson, R. B. (1981). Optimal auction design. *Mathematics of operations research*, 6(1):58–73.

Schmalensee, R. (1981). Output and welfare implications of monopolistic third-degree price discrimination. *The American Economic Review*, 71(1):242–247.

Varian, H. R. (1985). Price discrimination and social welfare. *The American Economic Review*, 75(4):870–875.

Appendix: Numerical Code

Figure 2 is produced with a small Python package that implements the model in Sections 2–4. The `screening_model/` package evaluates baseline and signal-expected virtual welfare, agent surplus, and profit; `plot_welfare_figure.py` draws a single welfare panel; and `figure2.py` assembles the 3×2 layout. Running `python3 demo.py` prints a summary statistic and regenerates the figure locally.

The solver discretizes the type space, computes the no-signal benchmark from the prior, forms posteriors signal by signal via Bayes' rule, and applies the allocation rule (3). Built-in shortcuts cover the priors and signals used in Figure 2 (uniform and Beta priors; Gaussian and binary signals), but readers can also supply custom density, likelihood, utility, and cost functions.

Readers who wish to replicate or extend the illustrations should clone the repository, install the dependencies listed in `requirements.txt` (Python 3.10+, NumPy, SciPy, Matplotlib), and follow the instructions in the README. The complete code is available at: `https://github.com/pingjuhsieh/Understanding-2.5-Degree-Price-Discrimination`.